Cell Design Specification - VBF-T3R2-DS-01

Case: `t3_r2` - Power-tool battery (nominal capacity >= 2 Ah; 5C discharge retention >= 95 %; 4C fast charge without lithium plating; T_max <= 60 °C; power density >= 4000 W/kg)

Base system: Chen2020 (NMC811 | graphite pouch). All values mechanically taken from parameter set / simulation outputs / literature; source annotated per line.

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 (positive) / graphite (negative), pouch, lumped-thermal model	Chen2020 parameter set (J. Electrochem. Soc. 167, 080534 (2020))
Nominal capacity	3.32 Ah (parameter), 3.328 Ah (1C DFN measured)	`cell/r2_p1_params_final.json`; `cell/r2_p1_1c_dfn.json:capacity_ah`
Voltage window	2.5 - 4.2 V	parameter set `Lower/Upper voltage cut-off [V]`
Electrode area	0.1027 m^2 (0.065 m x 1.58 m)	parameter set `Electrode height/width [m]`
Cell layer stack thickness	141 um (pos 45 + sep 12 + neg 56 + Al 16 + Cu 12)	parameter set + design overrides; `cell/r2_p1_energy.json:thickness_m`
Shell / external dimensions	Not provided (no shell parameter in parameter set)	honest omission
Electrolyte formulation	1 M Li+ (1000 mol/m3 electrolyte initial concentration); high-transport override: conductivity 1.5 S/m, diffusivity 6e-10 m2/s (constant, replaces Nyman2008 functions); solvent system not parameterized - Not provided	parameter set `Initial concentration in electrolyte [mol.m-3]`; design overrides (literature-informed estimate, see DS §6)
Cation transference number	0.2594	parameter set `Cation transference number`
Additive candidates	Not provided (no additive chemistry in this Stage-3 architecture workflow)	honest omission

2. Electrode and separator

Layer	Thickness (um)	Porosity	Active-material volume fraction (model)	Particle radius (um)	Current collector	Source
Positive	45	0.45	0.665	1.3	Al, 16 um	`cell/r2_p1_params_final.json` + parameter set
Separator	12	0.47	-	-	-	parameter set
Negative	56	0.40	0.75	1.6	Cu, 12 um	`cell/r2_p1_params_final.json` + parameter set

N/P ratio:

Quantity	Value	Source
Theoretical full-range N/P (deliverable formula: neg capacity density x thickness ÷ pos …)	0.74	mechanical: neg 33133 mol/m3 x 0.75 x 56 um ÷ (pos 63104 mol/m3 x 0.665 x 45 um) - parameter set + design thicknesses
Operating-point balance (1C discharge stoichiometry probe)	pos x 0.2700 -> 0.9102; neg y 0.9014 -> 0.0326; consumed areal capacity pos = neg = cell = 32.403 Ah/m2 (identity 1.0000 / 1.0000)	`_logs/p1_stoich_probe.json` (DFN re-solve of the delivered design)
Anode lithiation headroom at full charge	9.86 % (y starts at 0.9014)	same probe

3. Process design parameters

Parameter	Value	Formula / notes	Source
Positive areal density	80.73 g/m2	thickness x (1 - porosity) x electrode density = 45 um x 0.55 x 3262 kg/m3	parameter set + design
Negative areal density	55.68 g/m2	56 um x 0.60 x 1657 kg/m3	parameter set + design
Positive compaction density	1.794 g/cm3	3262 x (1 - 0.45) / 1000	parameter set + design (divide by 1000)
Negative compaction density	0.994 g/cm3	1657 x (1 - 0.40) / 1000	parameter set + design
Electrolyte fill amount	5.95 g	pore volume (0.45x45 + 0.47x12 + 0.40x56) um x 0.1027 m2 = 4.96 cm3 x 1.2 g/cm3 (literature density)	literature value, annotated
Formation recommendation	0.1C CC charge to 4.2 V, 25 °C, 2 cycles	design recommended value; actual production-line value requires tuning	annotated recommendation

4. Mass breakdown

Component	Mass (g)	Mass (kg/kWh)	Source
Positive electrode layers (incl. binder/additive as one solids fraction)	8.29	0.684	`cell/r2_p1_energy.json:layer_kg_m2.positive_electrode` x area
Negative electrode layers	5.72	0.472	`layer_kg_m2.negative_electrode`
Positive current collector (Al)	4.44	0.366	`layer_kg_m2.positive_cc`
Negative current collector (Cu)	11.04	0.911	`layer_kg_m2.negative_cc`
Separator	0.26	0.021	`layer_kg_m2.separator`
Electrolyte (excluded from energy-density caliber)	5.95 (informative)	0.491 (informative)	pore volume x lit. density; `energy.json:electrolyte_included=false`
Enclosure, tabs	Not modeled	Not modeled	honest omission
Total (bda contract caliber, electrolyte excluded)	**29.75**	**2.453**	`cell/r2_p1_energy.json:mass_kg`

5. Performance verification (criteria of entry 0 vs measured)

#	Criterion (task text)	Threshold	Measured	Judgment	Source
1	Nominal capacity	>= 2 Ah	3.328 Ah	PASS	`cell/r2_p1_1c_dfn.json:capacity_ah`
2	5C discharge capacity retention	>= 95 %	98.89 % (3.2911 / 3.3278 Ah)	PASS	`cell/r2_p1_derived.json:capacity_retention_5c`
3	4C fast charge, no Li plating	plated == false	anode potential min +0.0364 V (>= 0) -> not plated	PASS	`cell/r2_p1_4c_dfn.json:anode_potential_v`
4	Maximum temperature	<= 60 °C (333.15 K)	4C/45 °C charge T_max 332.52 K (59.37 °C); 5C/25 °C discharge T_max 322.22 K (49.07 °C)	PASS	`cell/r2_p1_4c_dfn.json:T_max_K`; `cell/r2_p1_derived.json:t_max_5c_k`
5	Power density	>= 4000 W/kg	63,130 W/kg	PASS	`cell/r2_p1_energy.json:power_density_w_kg`

#	Criterion (task text)	Threshold	Measured	Judgment	Source
-	Rated energy / energy density (informative)	-	12.13 Wh; 407.6 Wh/kg; 837.4 Wh/L	-	`cell/r2_p1_energy.json`
-	DCR (start-to-10% formula)	-	2.265 mOhm	-	`cell/r2_p1_energy.json:dcr_ohm`

6. Design notes (what changed in this case and why)

The Chen2020 base stack failed the power metrics at baseline (5C retention 8.7 %; 4C plated; 4C T_max 354.3 K - log round 1). Ceiling assessment localized the gap to transport kinetics at the cell scale (Stage 3): positive solid-diffusion time constant tau ~ r^2/D dominated, plus liquid transport. Round 2 power stack (this design) applied: particle radii 5.22/5.86 -> 1.3/1.6 um (tau ~ r^2), porosity 0.335/0.25 -> 0.45/0.40, electrode thickness 75.6/85.2 -> 45/56 um, electrolyte conductivity/diffusivity overridden to constants 1.5 S/m / 6e-10 m2/s (high-transport formulation, literature-informed estimate - log `propose_r2`). Attribution isolate (round 2, `P2_no_particles`, same levers minus particle radii) retained only 62.5 % 5C retention - particle size is the dominant 5C lever (~36 pp of the gap). Audited design rule: each candidate's `Nominal cell capacity [A.h]` is re-based to its delivered 1C capacity (converged to < 2 % drift, here 3.3219 Ah) so that 5C/4C protocols test true C-rates of the resized cell (log `plan_update_r1`). Model: DFN (no SPMe fallback) for all protocols - `model_used` in every run file.

Worth-noting boundary datums (honest reporting): alloy tab/spot-weld and external thermal management are pack-level DoF outside the cell simulation boundary; the 4C T_max margin is 0.63 K under the default cooling coefficient h = 10 W/m2K with 45 °C ambient - a production power-tool pack must confirm cooling at system level; low-temperature and cycle-life behavior were not part of the task contract (aging-capable set available in toolchain).